Mathematical modeling and application: in-course assignment

This assignment consists of two parts. In the first you will perform a programming task similar to those that appeared in the practicals. In the second you will answer questions related to the lectures. Please upload your answers to these parts as two separate documents: a `.ipynb` file for Part 1 and a `.word` or `.pdf` file for Part 2.

1 Part 1: Programming assignment

Please upload your solutions as a jupyter notebook (`.ipynb` file). **Do not upload an `.html` or `.pdf` file. If the file cannot be run you will not receive points.** Use separate code cells for each question. Annotate as much as you can with comments to explain your reasoning. If you make an error but your reasoning is sound you may still receive some points. Likewise, if your code errors, but you explain your reasoning you may still receive points.

Problem 1. Consider a cell that acquires a random number of mutations during a cell division. Now imagine that cell divides a random number of times within a fixed time interval. We are interested in the total number of mutations it has acquired at the end of the time period (see figure). Model this in the following manner:

- take the number of divisions D in the time interval T to be drawn from a Poisson distribution with rate parameter $\rho \cdot T$, where ρ is the cell's division rate.
- take the number of mutations M arising during a division to be drawn from a Poisson distribution with rate parameter μ, the cell's per-division mutation rate.
- The total number of mutations Z is the sum over D draws of M:

$$Z = M_1 + M_2 + \cdots + M_D \tag{1}$$

(a) Write a function that returns one realisation of Z. It should take as arguments the cell's division rate ρ, the cell's mutation rate μ, and the time interval T.

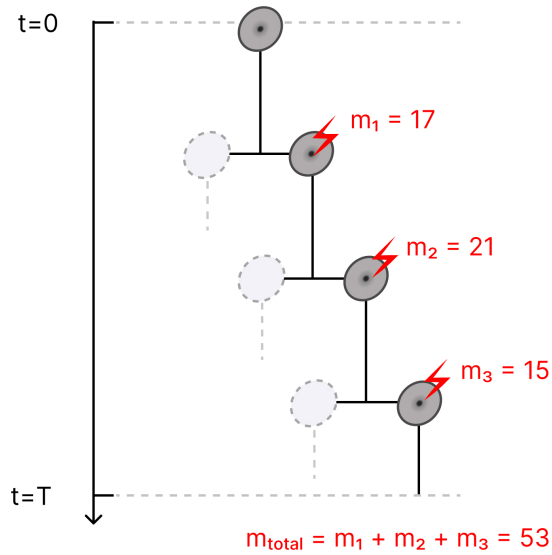

Figure 1: An example of the process where the cell undergoes 3 divisions and ends up with 53 mutations.

(b) Using the parameters $\rho = 2$ (*divisions per day*), $\mu = 20$ (*mutations per division*), and $T = 50$ (*days*), obtain 500 realisations of Z. Plot these in a histogram.

(c) Create a new function that takes the same arguments as the previously defined function, and returns the mean of 500 realisations of Z. Use this function to plot how the mean of Z varies with the time interval T.
TIP: you can use the numpy function `numpy.linspace(start,stop,num)` to generate `num` equidistantly spaced values between `start` and `stop`.

Problem 2. Consider a population of 100 stem cells in a niche somewhere in the body. Occasionally a stem cell will differentiate, leaving the population for good. Because the number of cells in this population is strictly regulated through hormonal signalling, when this occurs, a random cell in the population will divide to replace the differentiated cell. Thus the total stem pool maintains a fixed size $N = 100$ over time. Now imagine a mutant suddenly arises in the population and we are interested in its dynamics over time. What classical model can be used to describe this system?

(a) Create a function that takes as arguments:

- the current mutant population size
- the total population size

and returns the probability of a random cell chosen for differentiation being a mutant. Note that when choosing for differentiation, all cells have the same chance of being chosen.

Now Imagine the mutant bears a fitness advantage compared to the existing wildtype cells. Take the advantage to operate in the following manner: when a random cell is selected for ***division***, a cell bearing the mutation has a slightly larger chance to be chosen than a wildtype cell. In particular, with X and Y the sizes of the mutant and wildtype populations respectively, and $s > 1$ a positive real number denoting the strength of the fitness, a ***single*** mutant cell has a probability

$$P_{\text{single mutant cell}} = \frac{s}{Xs + Y} \tag{2}$$

of being selected, with $s > 1$. Note that if $s = 1$ this reduces to the neutral case.

(b) Create a function that takes the same arguments as in (a) and an additional argument s for the selective advantage. Have it return the probability that the cell chosen for ***division*** belongs to the mutant population.
Hints:

- Note that the probability of *any* mutant cell being chosen is

$$P_{\text{any mutant cell}} = XP_{\text{single mutant cell}} \tag{3}$$

- Since the total population is constant, we have $N = X + Y$ to be always true.

(c) Imagine the mutant arises at time $t_0 = 0$, and differentiations occur at random times. In particular, assume the time between successive differentiations in the population is exponentially distributed, with rate Nr_0 (where r_0 is the differentiation rate of a single cell). Construct a simulation evolved over a time T. At each differentiation event perform a differentiation and a division by randomly selecting cells according to the probabilities calculated in (a) and (b).

(d) Using parameter values $X_0 = 1$ (the initial size of the mutant is a single cell), $N = 200$ (the total population consists of 100 cells), $r_0 = 0.03$, $s = 2.3$, and $T = 200$, run the simulation 10 times and plot all mutant trajectories on the same figure. Be sure to show the correct values of t on the x-axis!

(e) Note that in some simulations the mutant population may still go extinct despite its selective advantage. Obtain an empirical estimation for the probability of extinction if we were to run the simulation for an infinite amount of time.

(f) Now consider the simulations where the mutant takes over the entire population. This is called fixation. Obtain an empirical estimate for the average time until fixation occurs.

2 Part 2: Written-answer questions

Please upload your answers to a Word (or pdf) document.

Problem 1. Luria and Delbrück designed a famous experiment to test whether resistance of bacteria to bacteria phages is pre-existing in the bacterial population, or bacteria can sense the presence of phages and resistance is a direct phenotypic response. Please describe briefly in your own words what the basic design and idea of the experiment is. What did Luria and Delbrück observe and what was the interpretation of these observations.

Problem 2. The majority of cancer genetic analysis today still rely on inferences of single sample bulk sequencing. A typical summary statistic of this data is the Site-frequency spectrum. Can you briefly describe what the typical expectation of a site-frequency spectrum of a neutrally evolving tumour is. How many peaks do we expect and what is the evolutionary explanation of these peaks. How would the Site-frequency spectrum change in the presence of a single strongly selected subclonal driver mutation?

Problem 3. Recently, the role of extrachromosomal DNA (ecDNA) in human cancers, e.g. in Glioblastoma has attracted new attention in the cancer research community. Single cell methods, such as single cell FISH or single cell ATTAC seq show extreme cell to cell ecDNA copy number variation. Some cells might carry one copy, other cells might carry hundreds of copies of ecDNA. Please explain in your own words, what is the evolutionary explanation for these observations and what are typical implications for cancer evolution and treatment response.